

CUCM → Microsoft Teams Phone

Enterprise Voice Migration Project Plan

Credit Union & Banking Hybrid Environment

Portfolio Project • Demonstration / Planning Artifact

1. Executive Summary

This project plan describes a controlled migration from Cisco Unified Communications Manager (CUCM) to Microsoft Teams Phone Enterprise Voice for a credit union or bank. The design assumes a regulated financial-services environment where voice availability, emergency calling, security, auditability, business continuity, and customer/member service are critical.

The migration uses a phased approach rather than a 'big bang' cutover. CUCM remains operational while Teams Phone is prepared, tested, and introduced by user/site wave. A Microsoft-certified Session Border Controller (SBC) can provide Direct Routing to existing PSTN services and can support controlled coexistence with remaining on-premises telephony during the transition. Microsoft documents Direct Routing as a way to connect Teams Phone to customer telephony infrastructure through a supported SBC and recommends validation of inbound/outbound calling before production rollout.

2. Project Objectives

- Migrate eligible CUCM users, phone numbers, calling policies, and selected voice applications to Teams Phone.
- Maintain business continuity for branches, contact-center-adjacent teams, lending, fraud, operations, and corporate users.
- Preserve required PSTN connectivity, caller ID, emergency calling, voicemail, hunt/queue workflows, and executive calling requirements.
- Establish secure and supportable Direct Routing using a Microsoft-certified SBC where the target architecture requires it.
- Use pilot and wave-based migration with measurable entry/exit criteria and a documented rollback path.
- Reduce reliance on legacy CUCM infrastructure while retaining temporary coexistence for dependencies that cannot immediately migrate.

3. Target-State Architecture

Reference architecture: Microsoft Teams Phone provides the user voice experience; Microsoft 365 provides identity and cloud administration; a certified SBC provides Direct Routing to the PSTN/carrier and, where required, controlled interworking with remaining CUCM/legacy voice resources.

Layer	Target Design	Key Considerations
User endpoint	Teams desktop/mobile; certified Teams phones where needed	Softphone-first; shared/common-area devices assessed separately
Voice platform	Microsoft Teams Phone	Licensing, calling policies, dial

		plans, voice routing
PSTN	Direct Routing through certified SBC	Carrier trunks, caller ID, emergency calling, redundancy
Legacy coexistence	CUCM retained during migration waves	Route only required ranges/dependencies; document ownership
Identity	Microsoft Entra ID / Microsoft 365	UPN, licensing, MFA, role-based administration
Network	Enterprise LAN/WAN + Internet connectivity	QoS, firewall, DNS, latency/jitter/loss, branch resiliency
Management	Teams Admin Center + Teams PowerShell + SBC tools	Configuration, monitoring, audit trail, automation
Security	Least privilege, certificates, segmentation, logging	Protect signaling, media, admin access, and sensitive calling data

4. Current-State Discovery — CUCM Assessment

Before configuring Teams, inventory the CUCM environment and classify every dependency.

Discovery Area	Information to Capture	Migration Decision
Users	DN, UPN, department, location, device type, manager	Migrate / retain / remediate
Numbers	DIDs, main numbers, toll-free, service numbers, ranges	Port, assign, retain, or route
Dial plan	Partitions, CSS, route patterns, translation patterns	Map to Teams dial plan/voice routes
PSTN	Carrier, SIP trunks, gateways, codecs, caller ID	Move via SBC or retain temporarily
Devices	Cisco desk phones, soft clients, analog endpoints	Replace / ATA / retain temporarily
Features	Hunt groups, pickup groups, forwarding, voicemail, paging	Map to Teams capabilities or third-party solution
Applications	IVR, recording, fax, alarms, elevator phones, branch devices	Dependency assessment before migration
Integrations	CRM, contact center, recording, compliance, directory	Validate vendor support and target architecture
Emergency calling	Locations, dispatchable addresses, carrier/service configuration	Validate before every production wave
Operations	CUCM admins, carrier contacts, NOC/help desk procedures	Update ownership and support model

5. Migration Strategy

Recommended sequence:

- Assess and document CUCM and PSTN dependencies.
- Define the Teams Phone target architecture and migration criteria.
- Select and size a Microsoft-certified SBC and confirm carrier/PSTN compatibility.

- Build Teams voice configuration while CUCM remains the production platform.
- Establish Direct Routing, number inventory, voice routing, emergency calling, and test identities.
- Create a small technical pilot and validate end-to-end calling.
- Migrate low-risk business users first, then expand by department/site.
- Migrate complex users and branch/operational scenarios only after dependencies are validated.
- Decommission CUCM components only after all required numbers, applications, devices, and integrations are removed from the dependency map.

Microsoft's current Direct Routing guidance calls for infrastructure/licensing review, a certified SBC, DNS/certificates/firewall and connectivity preparation, SBC connection, voice routing, user assignment, and validation. Microsoft also supports combining Direct Routing with other PSTN connectivity models where appropriate.

6. CUCM-to-Teams Coexistence Design

Scenario	Example During Migration	Control
Teams user → PSTN	Migrated employee places external call	Teams Phone → SBC → carrier
PSTN → Teams user	Carrier sends call for migrated DID	Carrier/SBC → Teams
Teams → CUCM user	Migrated employee calls remaining CUCM user	SBC/interworking route to CUCM
CUCM → Teams user	Legacy user calls migrated employee	CUCM route pattern → SBC → Teams
Main number	Branch/corporate main number remains on legacy platform temporarily	Explicit route and owner; test failover
Legacy analog	Alarm/fax/door/elevator/other endpoint	Retain or migrate only after validated replacement
Contact center	Existing contact-center number/workflow	Separate migration workstream; do not move blindly with users

The exact coexistence topology depends on the existing CUCM version, carrier topology, SBC capabilities, number ownership, and application dependencies. The project team should document the call-flow for every business-critical number before changing routing.

7. Teams Phone / Direct Routing Build

- Assign required Microsoft 365/Teams Phone licensing to pilot users.
- Register and verify the SBC FQDN in the Microsoft 365 tenant.
- Install a trusted public certificate and configure required DNS/firewall/connectivity.
- Connect the certified SBC to Teams Phone and verify the SBC connection.
- Configure dial plans, normalization rules, PSTN usages, voice routes, and voice routing policies.
- Assign Direct Routing numbers and voice routing policies to pilot users.
- Configure caller ID, emergency calling, voicemail, call forwarding, delegation, and relevant Teams calling policies.
- Implement SBC redundancy and carrier redundancy where the business continuity design requires it.
- Document configuration baselines and export relevant configuration before each production wave.

Microsoft's documented Direct Routing flow includes connecting and validating the SBC, enabling users, configuring call routing, and translating numbers as needed. Microsoft states that only certified SBCs are supported for Direct Routing.

8. Numbering & Dial-Plan Migration

Step	Activity	Acceptance Criteria
1	Export CUCM DNs/DIDs and carrier inventory	100% of numbers accounted for
2	Classify numbers by user/site/function	Every number has an owner and migration disposition
3	Normalize numbers to E.164 where appropriate	Inbound/outbound tests map correctly
4	Build Teams dial plan and voice routes	Test numbers route predictably
5	Prepare carrier/SBC routing changes	Carrier change ticket approved
6	Migrate pilot numbers	No critical call-flow defects
7	Migrate wave numbers	Wave acceptance signed
8	Validate main/service numbers	Business owner confirms behavior
9	Close port/routing work orders	Inventory and diagrams updated

Direct Routing supports phone numbers managed through the Teams tenant, while PSTN services and emergency-calling arrangements remain dependent on the organization's carrier/provider and deployment model.

9. Financial-Services Security & Compliance Workstream

- Use least-privilege administrative roles and separate day-to-day administration from privileged voice administration.
- Protect SBC management interfaces and restrict administrative access to approved management networks.
- Use trusted certificates, documented certificate renewal ownership, and change-controlled DNS.
- Document firewall rules and review them through the organization's security/change process.
- Define call-recording, voicemail, retention, legal-hold, and eDiscovery requirements with Compliance/Legal before enabling features.
- Classify any call recordings or transcripts that may contain member/customer information and apply approved retention/access controls.
- Document emergency-calling locations and validate dispatchable location behavior before each site goes live.
- Maintain auditable change records for number assignments, routing policy changes, carrier changes, and migration approvals.
- Coordinate with Information Security, Risk, Compliance, Legal, Network, Telecom, and Business Continuity teams.

10. Branch & Critical-Service Readiness

Critical Service	Pre-Go-Live Validation
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Branch main line	Inbound PSTN, auto attendant/queue behavior, transfer and overflow
Lending	Inbound/outbound calling, transfer, caller ID, voicemail
Fraud / security	Priority numbers, after-hours routing, escalation contacts
Cards / member services	Queue/agent routing and service continuity
Executive assistants	Delegation, shared lines, call pickup, mobile behavior
ATM/alarms/analog	Confirm whether device remains on CUCM/PSTN or has an approved replacement
Emergency calling	Address/location and emergency test process
Business continuity	WAN/Internet/carrier/SBC failure procedures

11. Sample 14-Week Project Schedule

Week	Workstream	Major Deliverables
1	Kickoff / governance	Charter, stakeholders, RACI, change process
2	CUCM discovery	Number/device/application inventory
3	Architecture	Target design, SBC sizing, carrier plan
4	Network/security	Firewall, DNS, certificates, QoS readiness
5	Teams build	Licensing, policies, dial plans, test accounts
6	SBC / PSTN	Direct Routing connection and carrier validation
7	Coexistence	CUCM ↔ Teams routing and dependency validation
8	Pilot	Pilot users, devices, numbers, test execution
9	Pilot remediation	Defect fixes and operational runbook
10	Wave 1	Low-risk departments/site
11	Wave 2	Additional departments/site
12	Wave 3	Complex users / branches
13	Final migration	Remaining eligible users/numbers
14	Stabilization / retirement	Hypercare, inventory reconciliation, CUCM retirement decision

12. Migration Wave Entry / Exit Criteria

Stage	Entry Criteria	Exit Criteria
Pilot	Architecture approved; test users ready; rollback tested	Critical tests pass; no Sev-1/Sev-2 defects
Wave 1	Pilot accepted; help desk ready	Business owner signs acceptance
Wave 2+	Prior wave stable; dependencies	All critical numbers/functions

	cleared	validated
Final	Remaining inventory reconciled; carrier changes approved	Migration completion signed
CUCM retirement	No production dependencies remain	Decommission/change record approved

13. Cutover Runbook

- Freeze nonessential voice configuration changes.
- Confirm approved change ticket, implementation team, carrier/SBC contacts, and escalation bridge.
- Back up/export CUCM, SBC, and relevant Teams configuration baselines.
- Confirm monitoring and test endpoints are available.
- Apply carrier/SBC routing changes for the approved number range.
- Assign Teams Phone numbers and voice routing policies to the migration wave.
- Validate inbound PSTN, outbound PSTN, Teams-to-Teams, Teams-to-CUCM, CUCM-to-Teams, transfer, voicemail, caller ID, and emergency calling scenarios as applicable.
- Obtain business-owner acceptance.
- Monitor call quality, failed calls, carrier alarms, SBC health, and service desk tickets during hypercare.
- Close the change only after the agreed stabilization criteria are met.

14. Rollback Strategy

- Define rollback triggers before cutover—for example, widespread failed calling, emergency-calling failure, critical business-number failure, or unacceptable service impact.
- Reverse the carrier/SBC routing change for the affected range or wave.
- Restore CUCM routing/assignments as documented in the pre-change baseline.
- Reassign affected users/numbers to the previous service where technically supported.
- Validate inbound/outbound calling and critical business workflows.
- Communicate rollback status and open a problem-management record for root-cause analysis.
- Do not retire CUCM or remove legacy dependencies until the rollback window and stabilization period have expired.

15. Sample Test Plan & Results

ID	Test Case	Expected Result	Sample Result
CU-001	Teams user → external PSTN	Call connects; correct caller ID	PASS
CU-002	External PSTN → Teams DID	Call rings correct Teams user	PASS
CU-003	Teams → remaining CUCM user	Call reaches CUCM endpoint	PASS
CU-004	CUCM → migrated Teams user	Call reaches Teams endpoint	PASS
CU-005	Internal transfer	Blind/consult transfer completes	PASS
CU-006	Voicemail	Unanswered call reaches mailbox; notification	PASS

		works	
CU-007	Caller ID	Approved outbound identity displayed	PASS
CU-008	Main number / auto attendant	Business-approved routing and overflow work	PASS
CU-009	Emergency calling	Approved test procedure confirms location/routing	PASS
CU-010	SBC failover	Calls continue through backup path where designed	PASS
CU-011	Carrier/PSTN failure simulation	Documented business-continuity behavior occurs	PASS
CU-012	CUCM rollback	Legacy service restored within target window	PASS
CU-013	Call quality	Audio is intelligible; QoS metrics meet project threshold	PASS
CU-014	Mobile Teams calling	Approved mobile calling scenario works	PASS
CU-015	Wave inventory reconciliation	All migrated numbers/users match approved list	PASS

Note: These are sample demonstration results for a portfolio project, not evidence of production testing.

16. Defect Severity Model

Severity	Definition	Migration Action
Sev 1	Critical service unavailable, emergency calling failure, widespread PSTN outage	Stop/rollback immediately
Sev 2	Major business function impaired for multiple users	Do not advance wave until resolved or formally waived
Sev 3	Limited functional issue with workaround	Track and remediate during hypercare
Sev 4	Cosmetic/documentation/minor issue	Backlog with owner/date

17. Project Roles / RACI

Role	Primary Responsibility
Project Manager	Schedule, risks, dependencies, change governance
Unified Communications Engineer	CUCM/Teams architecture, routing, SBC, migration execution
Microsoft 365 Engineer	Teams Phone, identity, licensing, policies, PowerShell
Cisco/CUCM Engineer	Current-state CUCM, dial plan, legacy routing, rollback

Network Engineer	QoS, firewall, DNS, WAN/Internet readiness
Security Engineer	Certificates, segmentation, privileged access, security review
Carrier / SBC Engineer	PSTN trunks, SBC configuration, number routing
Service Desk	User communications, incident triage, knowledge articles
Business Owner	UAT, acceptance, business-process validation
Compliance / Risk / Legal	Retention, recording, regulatory and risk requirements

18. Risks & Mitigations

Risk	Impact	Mitigation
Incomplete CUCM inventory	Unexpected call failures	Discovery workbook + business-owner validation
Unsupported legacy device	Critical endpoint outage	Classify devices before migration; retain or replace
Incorrect dial plan	Misrouted calls	Build test matrix; validate normalization and routes
SBC/carrier failure	PSTN outage	Redundancy, monitoring, carrier escalation plan
Emergency calling misconfiguration	High safety/regulatory risk	Validate location data before every site/wave
Network quality issues	Poor call quality	Readiness testing, QoS, monitoring, remediation
Unplanned application dependency	Business disruption	Application dependency map and owner sign-off
User adoption issues	Support volume	Training, quick-reference guides, hypercare
Premature CUCM retirement	Rollback impossible	Retain legacy platform until exit criteria are met

19. Deliverables

- Current-state CUCM inventory and dependency workbook
- Target Teams Phone architecture diagram
- SBC/PSTN design and carrier implementation plan
- Number-porting/number-assignment matrix
- Teams voice routing and dial-plan design
- Security and compliance assessment
- Pilot and migration-wave runbooks
- UAT/test plan and results
- Rollback plan
- Help-desk knowledge article and user communications
- As-built documentation and updated network/voice diagrams
- CUCM retirement/decommission plan

20. Portfolio / Interview Talking Points

This project can be presented as a demonstration architecture and planning artifact. A concise interview description:

“I developed a phased CUCM-to-Microsoft Teams Phone migration strategy for a financial-services environment. The approach starts with CUCM discovery and dependency mapping, then establishes Teams Phone and Direct Routing through a certified SBC, builds coexistence call flows, pilots a controlled user group, migrates users in waves, and uses measurable test/rollback criteria before retiring legacy infrastructure. I also incorporated emergency calling, security, carrier dependencies, branch readiness, and business continuity into the migration plan.”

21. Reference Architecture Notes

Microsoft's Direct Routing documentation states that Direct Routing connects customer telephony infrastructure to Teams Phone through a supported SBC and describes infrastructure, licensing, networking, certificate, routing, and validation requirements. Microsoft also documents voice routes, PSTN usages, voice routing policies, and active/backup SBC routing concepts. These references should be rechecked against the current Microsoft documentation and the selected SBC/carrier design before a real production implementation.

- Microsoft Learn — Plan Direct Routing: <https://learn.microsoft.com/en-us/microsoftteams/direct-routing-plan>
- Microsoft Learn — Connect your SBC to Direct Routing: <https://learn.microsoft.com/en-us/microsoftteams/direct-routing-connect-the-sbc>
- Microsoft Learn — Configure call routing for Direct Routing: <https://learn.microsoft.com/en-us/microsoftteams/direct-routing-voice-routing>
- Microsoft Learn — PSTN connectivity options: <https://learn.microsoft.com/en-us/MicrosoftTeams/pstn-connectivity>

22. Portfolio Disclaimer

This document is a professional demonstration project plan designed to show migration planning, voice engineering, testing, documentation, and change-management skills. Unless separately documented, sample test results and implementation details should not be represented as production work performed for a specific financial institution.